

NAME :- U.PRANEETH

ROLL NO:-2520030188

SEC :-03

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1      -- 1. Setup database without triggering drop warnings on the first run
2 •    CREATE DATABASE IF NOT EXISTS set_ops_db;
3 •    USE set_ops_db;
```

	#	Time	Action
	1	11:20:08	CREATE DATABASE IF NOT EXISTS set_ops_db
	2	11:20:08	USE set_ops_db
	3	11:20:08	DROP TABLE IF EXISTS table_A, table_B

```
6 •    DROP TABLE IF EXISTS table_A, table_B;
7
8      -- 2. Create two simple tables for comparison
9 •    CREATE TABLE table_A (id INT, item_name VARCHAR(50));
10 •   CREATE TABLE table_B (id INT, item_name VARCHAR(50));
11
12      -- 3. Insert overlapping data
13      -- (Apple is unique to A, Date is unique to B, Banana and Cherry are s
14 •   INSERT INTO table_A VALUES (1, 'Apple'), (2, 'Banana'), (3, 'Cherry');
15 •   INSERT INTO table_B VALUES (2, 'Banana'), (3, 'Cherry'), (4, 'Date');
16
```

	4	11:20:09	CREATE TABLE table_A (id INT, item_name VARCHAR(50))
	5	11:20:09	CREATE TABLE table_B (id INT, item_name VARCHAR(50))
	6	11:20:09	INSERT INTO table_A VALUES (1, 'Apple'), (2, 'Banana'), (3, 'Cherry')
	7	11:20:09	INSERT INTO table_B VALUES (2, 'Banana'), (3, 'Cherry'), (4, 'Date')

```
19  |-- 1. UNION (Combines both tables, removes duplicates)
20  -- Expected Output: Apple, Banana, Cherry, Date
21  • SELECT * FROM table_A
22  UNION
23  SELECT * FROM table_B;
24
25  -- 2. UNION ALL (Combines both tables, keeps duplicates)
26  -- Expected Output: Apple, Banana, Cherry, Banana, Cherry, Date
27  • SELECT * FROM table_A
28  UNION ALL
29  SELECT * FROM table_B;
30
31  -- 3. INTERSECT (MySQL Workaround using INNER JOIN)
32  -- Returns only the rows that exist in BOTH tables.
33  -- Expected Output: Banana, Cherry
34  • SELECT A.*
35  FROM table_A A
36  INNER JOIN table_B B
37  ON A.id = B.id AND A.item_name = B.item_name;
38
```

ip

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	id	item_name			
▶	1	Apple			
	2	Banana			
	3	Cherry			
	4	Date			

Result 7 ×

Output

Action Output

	#	Time	Action
✓	10	11:20:09	SELECT A.* FROM table_A A INNER JOIN table_B B ON A.id = B.id AND A.item_name = B.item_name
✓	11	11:20:10	SELECT A.* FROM table_A A LEFT JOIN table_B B ON A.id = B.id AND A.item_name = B.item_name
⚠	12	11:20:19	CREATE DATABASE IF NOT EXISTS set_ops_db
✓	13	11:23:15	SELECT * FROM table_A UNION SELECT * FROM table_B
✓	14	11:23:18	SELECT * FROM table_A UNION SELECT * FROM table_B
✓	15	11:23:20	SELECT * FROM table_A UNION SELECT * FROM table_B

38

39 | -- 4. MINUS / EXCEPT (MySQL Workaround using LEFT JOIN)

40 | -- Returns rows that exist in table_A but DO NOT exist in table_B.

41 | -- Expected Output: Apple

42 • SELECT A.*

43 FROM table_A A

44 LEFT JOIN table_B B

45 ON A.id = B.id AND A.item_name = B.item_name

46 WHERE B.id IS NULL;

Result Grid   Filter Rows: Export:  Wrap Cell Content: 

	id	item_name
▶	1	Apple

Result 12 ×

Output

 Action Output ▼

	#	Time	Action
✓	15	11:23:20	SELECT * FROM table_A UNION SELECT * FROM table_B
✓	16	11:24:32	SELECT A.* FROM table_A A LEFT JOIN table_B B ON A.id = B.id AND A.item_name
✓	17	11:24:36	SELECT A.* FROM table_A A LEFT JOIN table_B B ON A.id = B.id AND A.item_name
✓	18	11:24:39	SELECT A.* FROM table_A A LEFT JOIN table_B B ON A.id = B.id AND A.item_name
✓	19	11:24:41	SELECT A.* FROM table_A A LEFT JOIN table_B B ON A.id = B.id AND A.item_name
✓	20	11:24:44	SELECT A.* FROM table_A A LEFT JOIN table_B B ON A.id = B.id AND A.item_name